

PROJECT PROPOSAL

Project Title

Disaster Relief Management System (DRMS)

Submitted By

Name: Roni Parvez

Name: Shahera Parvin Bristy

Student ID: 42250102284

Student ID: 42250303024

Department: Computer Science & Engineering (CSE)

Course: Database Management System (DBMS)

University: Northern University Bangladesh

Semester: 3rd **Section :** E

Submission Date: 26/06/2026

1. Introduction

Natural disasters such as floods, cyclones, earthquakes, landslides, and fires often cause severe damage to human lives, infrastructure, and the environment. During emergency situations, relief operations frequently face major challenges due to poor coordination, lack of centralized information, inefficient resource allocation, and delayed communication among government agencies, NGOs, volunteers, and affected communities.

The proposed **Disaster Relief Management System (DRMS)** is a centralized, database-driven platform designed to efficiently manage disaster-related information. The system will support victim registration, volunteer management, shelter allocation, relief inventory management, donation tracking, and relief distribution. Its main purpose is to improve coordination among stakeholders and ensure faster, more transparent, and more effective disaster response operations.

2. Problem Statement

Current disaster relief operations are often handled manually or through disconnected systems, which leads to several critical problems:

- Inefficient victim registration
- Poor relief distribution tracking
- Resource shortages and duplication
- Lack of real-time inventory monitoring
- Difficulty in managing volunteers
- Limited transparency in donation utilization
- Weak reporting and decision-making support

These issues delay emergency response and reduce the overall effectiveness of humanitarian assistance.

3. Objectives

The primary objective of this project is to develop a centralized database management system that supports efficient disaster relief operations.

Specific Objectives

- Develop a secure disaster information management system.
 - Maintain victim records efficiently.
 - Manage shelters and accommodation capacity.
 - Track relief inventory in real time.
 - Manage volunteers and their assignments.
 - Record donations and monitor fund utilization.
 - Ensure transparent relief distribution.
 - Generate analytical reports for decision-makers.
 - Improve data accuracy and reduce redundancy.
-

4. Scope of the Project

The system will support the following modules:

- User Authentication & Role Management
- Disaster Registration

- Victim Registration
 - Shelter Management
 - Volunteer Management
 - Relief Inventory Management
 - Donation Management
 - Relief Distribution Tracking
 - Missing Person Management
 - Medical Assistance Records
 - Reporting & Analytics
 - Activity Logging
-

5. Proposed Features

User Module

- Login
- Registration
- Role-based Access Control

Disaster Module

- Disaster Type
- Disaster Location
- Severity Level
- Status Management

Victim Module

- Personal Information
- Family Information
- Injury Status
- Required Assistance

Shelter Module

- Shelter Capacity
- Available Beds
- Occupancy Status
- Shelter Facilities

Volunteer Module

- Registration

- Skills
- Availability
- Assignment

Relief Inventory Module

- Food
- Water
- Medicine
- Clothes
- Emergency Equipment

Donation Module

- Donor Information
- Cash Donation
- Goods Donation
- Donation History

Distribution Module

- Relief Request
- Distribution Records
- Delivery Status

Reporting Module

- Disaster Reports
 - Victim Statistics
 - Inventory Reports
 - Volunteer Reports
 - Donation Reports
-

6. Expected Database Entities

The database will consist of approximately **20–25 normalized tables**, including:

- Users
- Roles
- Disasters
- DisasterTypes
- Victims
- Families

- Shelters
 - ShelterResidents
 - Volunteers
 - VolunteerAssignments
 - ReliefItems
 - Inventory
 - Suppliers
 - Donations
 - Donors
 - Distribution
 - DistributionDetails
 - MissingPersons
 - MedicalAssistance
 - Vehicles
 - EmergencyContacts
 - ActivityLogs
-

7. Functional Requirements

The system shall:

- Register disasters.
 - Register victims.
 - Assign shelters.
 - Assign volunteers.
 - Manage relief inventory.
 - Track donations.
 - Record relief distribution.
 - Search disaster records.
 - Generate reports.
 - Maintain historical records.
-

8. Non-Functional Requirements

- Secure authentication
- High data integrity

- Fast query execution
 - User-friendly interface
 - Data consistency
 - Scalability
 - Backup and recovery support
-

9. Database Technologies

The project will utilize the following technologies and concepts:

- **Database:** MySQL
 - **Query Language:** SQL
 - **ER Diagram**
 - **Normalization** up to 3NF
 - **Views**
 - **Stored Procedures**
 - **Triggers**
 - **Transactions**
 - **Indexing**
 - **Foreign Keys**
 - **Constraints**
-

10. Expected Outcomes

The proposed system will:

- Improve disaster response efficiency.
 - Reduce data redundancy.
 - Increase transparency.
 - Improve inventory management.
 - Enable faster decision-making.
 - Enhance coordination among stakeholders.
 - Provide reliable reporting and analytics.
-

11. Future Enhancements

The system can later be extended with the following features:

- Mobile Application
 - GIS/Map Integration
 - SMS Notifications
 - Email Alerts
 - QR Code-Based Relief Distribution
 - AI-Based Resource Prediction
 - Real-Time Disaster Monitoring
 - Weather API Integration
-

12. Conclusion

The **Disaster Relief Management System** is designed to provide a reliable, centralized, and efficient database solution for disaster response management. By integrating victim management, shelter allocation, volunteer coordination, inventory control, donation tracking, and reporting into a single platform, the proposed system will significantly improve the effectiveness and transparency of disaster relief operations. This project demonstrates the practical application of Database Management System concepts in solving real-world humanitarian challenges.

Prepared By

Roni Parvez & Shahera Parvin Bristy

Department of Computer Science & Engineering

Northern University Bangladesh